

Alexander Brown

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Professional Summary

Computer Science M.S. candidate at Georgia Tech specializing in machine learning, with hands-on experience building AI and machine learning systems across computer vision, reinforcement learning, and full-stack development. Comfortable moving fast across the stack, picking up new tools quickly, and applying ML to real-world problems.

Education

Georgia Institute of Technology

August 2025 - Expected May 2027

M.S. Computer Science, Machine Learning Specialization

Relevant Coursework: Machine Learning, Computer Networks, Robotics: AI Techniques, Machine Learning for Trading
GPA: 4.0

University of California, Irvine

September 2022 - June 2025

B.S. Computer Science, Intelligent Systems Specialization

Relevant Coursework: Artificial Intelligence, Databases, Systems, Data Structures, Algorithm Analysis, Information Retrieval, Embedded Systems, Graph Algorithms, Probability, Statistics
GPA: 3.85

Research

Graduate Research Assistant

May 2026 - Present

Georgia Tech Human Augmented Analytics Group (HAAG)

- Build computer vision pipelines to detect and classify Hume's Leaf Warblers from Himalayan camera trap footage, training on a dataset of 50,000+ annotated frames including 13,000+ positive examples
- Evaluate YOLO and DETR/RT-DETR detection approaches across annotation-limited ecological data, contributing to model selection for an end-of-summer publication
- Investigate tracking and behavior analysis methods to support scalable, automated wildlife monitoring

Machine Learning Research Assistant

September 2024 - April 2025

California Institute of Telecommunications & Information Technology (Cal IT2)

- Developed GRU, LSTM, RNN, and RBF-RNN time-series models in PyTorch to simulate thermal energy storage system behavior from sensor data, reducing inference time to under 1 second vs. a 10-minute traditional simulation runtime

Work Experience

Full-Stack Web Developer

March 2023 - June 2025

University of California, Irvine, Student Affairs Information Technology

- Developed and maintained full-stack web applications using Laravel, PHP, JavaScript, CSS, and MySQL, supporting 9,000+ annual users across student-facing UCI systems
- Automated QA workflows using Cypress testing and custom scripts, improving release reliability

Projects

AI BWC Footage Tagging and Purging

January 2025 - June 2025

Capstone Project Team Lead

- Awarded 1st Place Computer Science Winner at UCI ICS Expo
- Built AI-driven computer vision software for the Signal Hill Police Department to identify and categorize body-worn camera footage
- Implemented ensemble ML models incorporating ImageBind, EfficientNet, and transformer architectures to classify 16 terabytes of untagged footage

Reinforcement Learning Agent: Cuphead

January 2025 - March 2025

AI Project

- Built a reinforcement learning agent to defeat Cuphead's Root Pack boss by combining YOLO-based real-time object detection with a Deep Q-Network (DQN) in a fast-paced, visually complex environment

Skills

Languages & Tools: Python, SQL, Git, Docker, AWS, Claude Code, GPT Codex, AI coding agents, Bash, Java, C++

Frameworks & Libraries: PyTorch, Hugging Face, Scikit-learn, NumPy, pandas, Matplotlib

Techniques: Computer Vision, Machine Learning, Reinforcement Learning, CI/CD, End-to-End Testing